MAE 3724: Dynamical Systems and Intro To Control

Homework-7

Bonus problems are optional

1. **Sinusoidal Steady-State Response of LTI Systems** **[20 marks]**

   Consider a linear time-invariant (LTI) system with transfer function

   $$G(s) = \frac{Y(s)}{U(s)}.$$

   Prove that if the input signal is

   $$u(t) = A_0 \sin(\omega t),$$

   then the steady-state output can be expressed as

   $$y(t) = K A_0 \sin(\omega t + \phi),$$

   where

   $$K = |G(j\omega)|, \qquad (1)$$
   $$\phi = \arg\left(G(j\omega)\right). \qquad (2)$$

   Hint: You may follow the proof sketch discussed in class.

   Bonus: 10 marks: How may you estimate a sufficiently long time for the system to show steady-state behavior?

2. **Bode Plots** **[40 marks]**

   (a) **[10 marks]** Derive relationships to convert slopes between the following units:

   $$\text{dB/decade}, \quad \text{dB/century}, \quad \text{dB/octave}.$$

   (b) **[20 marks]** Sketch the asymptotic Bode magnitude plot for the transfer function

   $$G(s) = \frac{50(s+2)}{s(s+10)}.$$

   Clearly show the individual contributions of each first-order term, as well as the overall magnitude plot.

   (c) **[10 marks]** Generate the exact Bode magnitude plot using MATLAB and compare it with your asymptotic approximation. Briefly comment on any differences observed.

   (d) **[Bonus: 10 marks]** Sketch the asymptotic phase plot and compare it with the MATLAB-generated phase response.

3. **Frequency Response of Second-Order Systems** **[40 marks]**

   Consider the following second-order transfer functions:

   $$G_1(s) = \frac{100}{s^2 + 4s + 100}$$

   $$G_2(s) = \frac{100}{s^2 + 16s + 100}$$

   For **each** system, answer the following:

(a) **[10 marks]** Determine the natural frequency, damped natural frequency, and resonant frequency.

(b) **[10 marks]** Compute the magnitude (in dB) at the natural frequency, damped natural frequency, and resonant frequency.

(c) **[10 marks]** Determine the phase angle at the natural frequency, damped natural frequency, and resonant frequency.

(d) **[10 marks]** Determine the bandwidth of the systems.

Figure 1: You may need this for problem 2

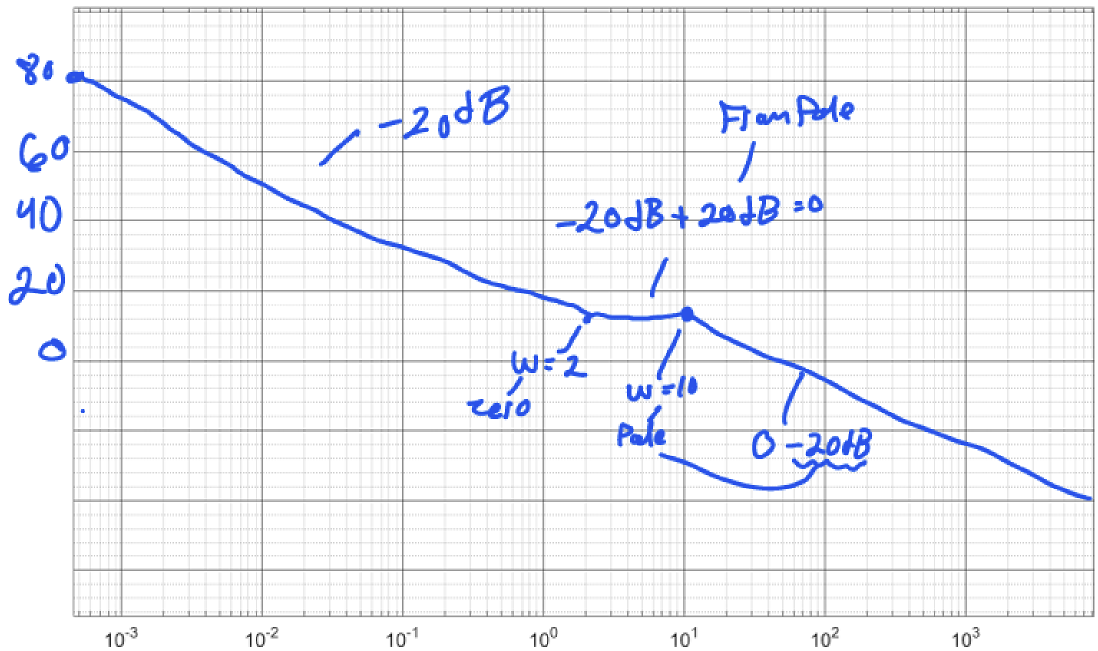

Figure 1: You may need this for problem 2

Starting Point = $20\log_{10}\left(\frac{gain}{w^n}\right)$ ← where $1/s^n$

$20\log_{10}\left(\frac{10}{10^{-3}}\right) = 80dB$

$20\log_{10}\left(\frac{10}{2}\right) = 14dB$

Valid til first Pole/zero